

Professional Summary

Simone Facchiano is a PhD student in the **National PhD Program in Artificial Intelligence** at Sapienza University of Rome. His research focuses on **computer vision and the interpretability** of large generative models, with additional experience in audio generation. He is the first author of **peer-reviewed publications** in these areas. His doctoral scholarship is funded by **CINECA**. Starting from February 2026, he will also serve as a Visiting Researcher at the **Max Planck Institute for Informatics** in Saarbrücken.

Simone earned his Bachelor's degree in **Statistics** with honors (110 cum laude), followed by a Master's degree in **Data Science**, also with honors (110 cum laude). He completed the university's **Honours Program**, and graduated ranking within the **top 2%** of his cohort. During this period, he conducted research on video generation using Mamba and Diffusion Models.

Throughout his academic journey, Simone was awarded the *Certificate of Merit* by the National Association of Carabinieri. He was part of the winning team at the **AWS Training Camp Jam Session 2023** at Sapienza University and led the winning team in the Statistical Learning 2023 Hackathon. He also spent a semester abroad at Maastricht University as part of the Erasmus exchange program.

Ph.D. Activity

- **Ph.D. in Artificial Intelligence** [ROME, SAARBRUCKEN] *November 2024 – Present*
 - First author of the papers “**Video Unlearning via Low-Rank Refusal Vector**” (in collaboration with NVIDIA) and “**Activation Patching for Interpretable Steering in Music Generation**”, both under submission at top-tier AI conferences.
 - Teaching experience as **Lecturer** and **Teaching Assistant** for *Fundamentals of Data Science* and *Advanced Machine Learning* (including lectures and hands-on labs).
 - Supervised and mentored several **Master's and Bachelor's students**, including participants in **Honours Programs**, on AI and deep learning research projects.
 - Active contributor to the AI research community, including volunteering at **ICIAP 2025** and attending the **ICVSS 2025** summer school.

Education

- **Master's Degree in Data Science** [ROME, MAASTRICHT] *September 2022 – October 2024*
 - Graduated with honors (110 cum laude), completing a master's thesis titled “*Llama for User Support: Developing a Large Language Model as a Ticketing System*” in collaboration with **CINECA**. This work involved extensive use of the **Leonardo** supercomputer, providing hands-on experience with large-scale **HPC** workflows.
 - Selected for the university's **Honours Program**, and graduated within the **top 2%** of his cohort. During this period, he conducted research on **State Space Models** applied to video generation with diffusion models, exploring them as a more efficient and scalable alternative to attention for long-sequence processing.
 - **Erasmus exchange** at *Maastricht University* during the 2023 winter semester.
 - Winner and team leader of both the “Statistical Learning 2023 Hackathon” and the “**Amazon Web Services (AWS) Training Camp Jam Session 2023**” at Sapienza University.
 - Recognized for academic merit through inclusion in the university's merit list for 2022–2023.
 - Served as *Tutor of Statistical Inference* during the 2024 winter exam session for the Bachelor's program in Management Statistics at Sapienza University.
- **Bachelor's Degree in Management Statistics** [ROME] *September 2019 – July 2022*
 - Graduated with honors (110 cum laude), completing a data-driven thesis on “*Access to Food in Cambodia, Myanmar, and Vietnam*” with data provided by the World Bank.
 - Consistently recognized for strong academic performance, with inclusion in the university's merit list for two consecutive years (2020–2021, 2021–2022).
 - Awarded the Certificate of Merit by the National Association of Carabinieri for “outstanding academic achievements”.